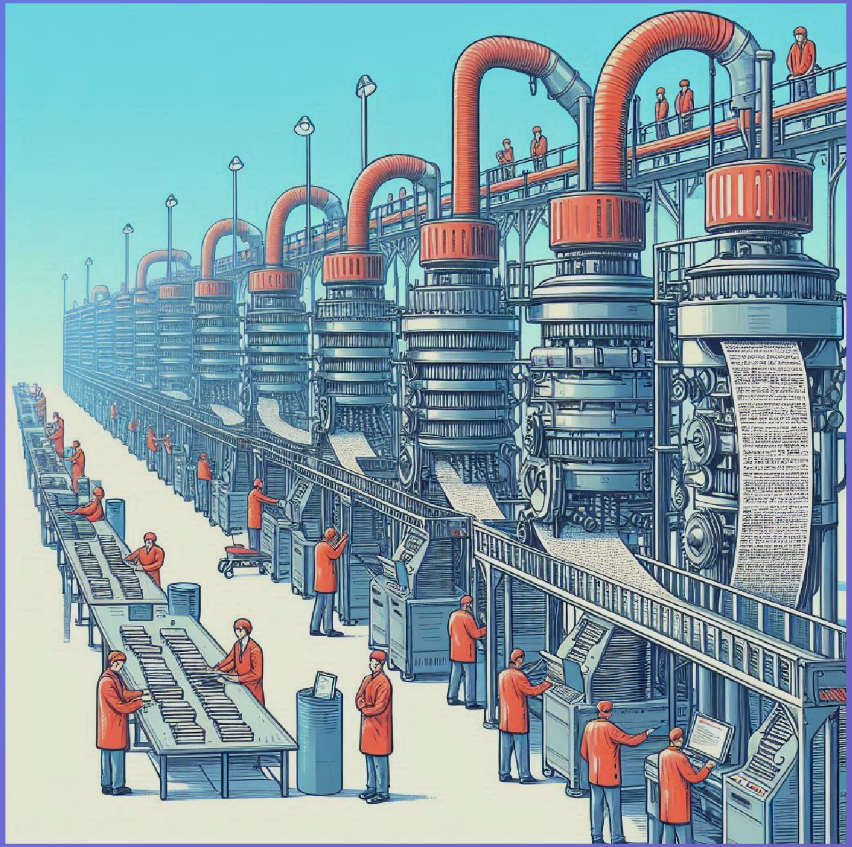


Marp template slides demo

Course Name

Oscar José Pellicer Valero

`Oscar.Pellicer@uv.es`

Departament d'Enginyeria Electrònica
Escola Tècnica Superior d'Enginyeria

This template demonstrates all the features available in Marp + the `extra.css` theme

Font and text styling

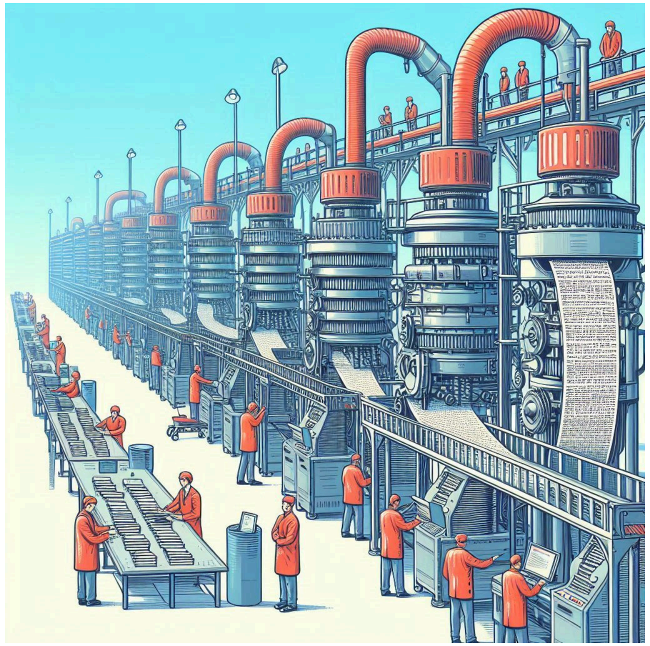
This slide demonstrates the various text styling options:

- **Bold text** for emphasis
- *Italic text* for highlights
- `Code snippets` for technical content
- Any ***combination*** ***`of the above`***
- Links: Google
- Mathematical Expressions:
  - Inline math: $P(w|h) = \frac{C(h,w)}{C(h)}$
  - Block math:

$$P(w_1, \ldots, w_n) = P(w_1)P(w_2|w_1)P(w_3|w_1, w_2)\ldots P(w_n|w_1, \ldots, w_{n-1})$$

> Citations

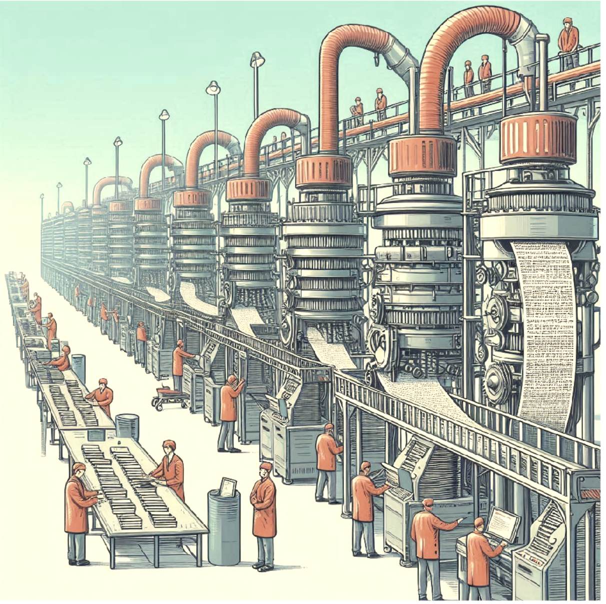
Centered images



This demonstrates centered images using the `center` attribute. You can also control the size with the `w:` parameter (or with the `h:` parameter for height):

```
![center w:400px](imgs/your-image.png)
```

Right and left floating images

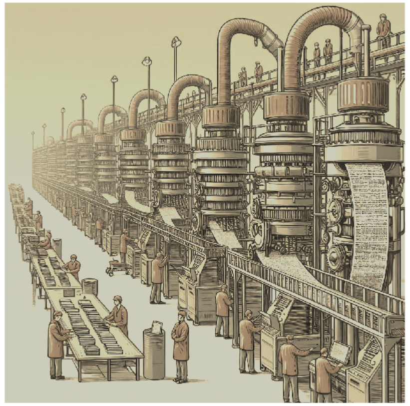


This text flows around the left-aligned image. The image is positioned using the `left` attribute in the alt text, and the width is controlled with the `w:` parameter.

The text continues to wrap around the image naturally, creating a professional layout that's commonly used in academic presentations.

We can break the floating with the `clear` class:

```
<div class="clear"></div>
```



Similarly, this text flows around a right-aligned image. The positioning is controlled through the alt text attributes, making it easy to create dynamic layouts without complex CSS.

Notice that this slide has the `small` class (`<!-- _class: small -->`), so the text is smaller than the default.

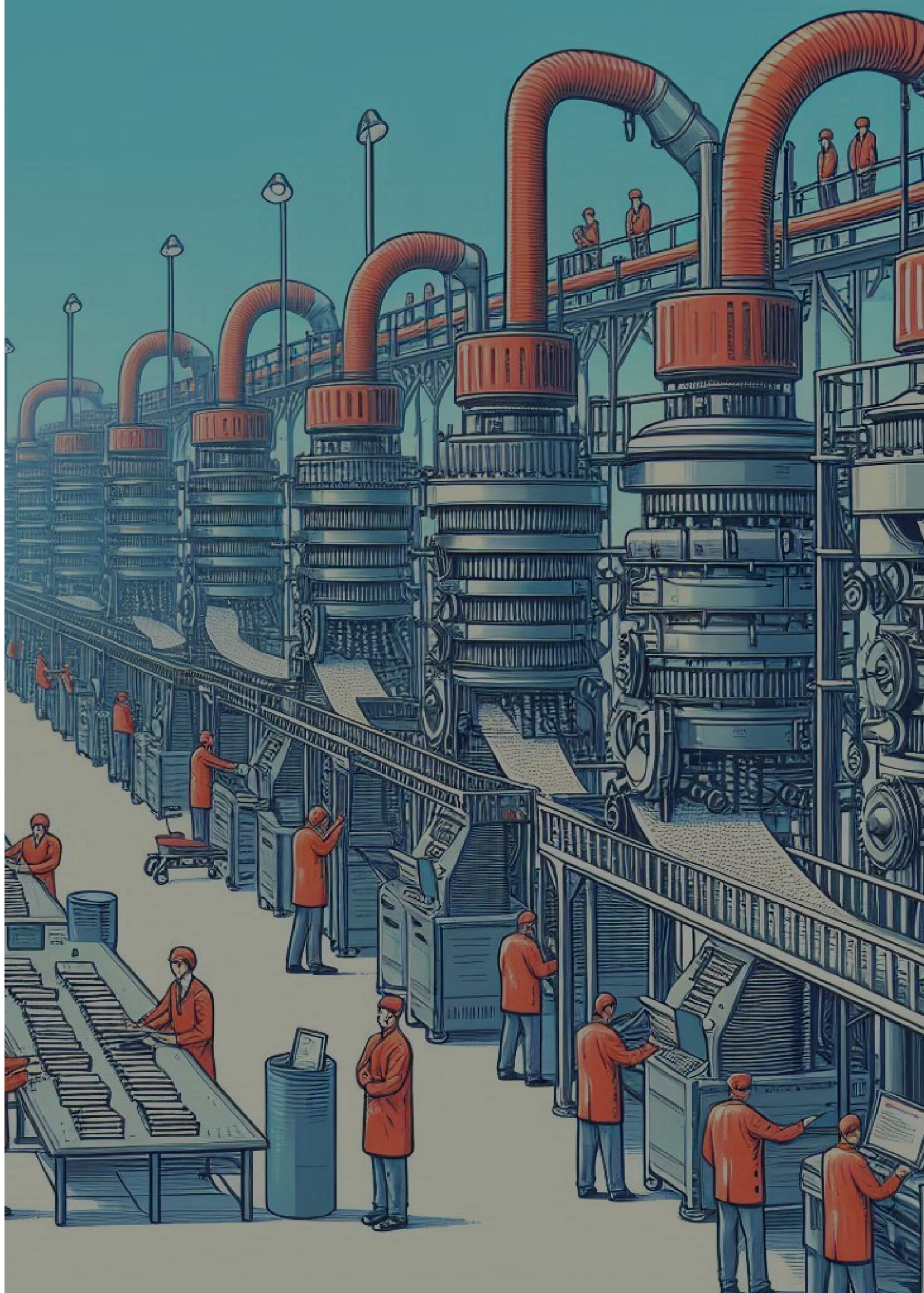
Background images

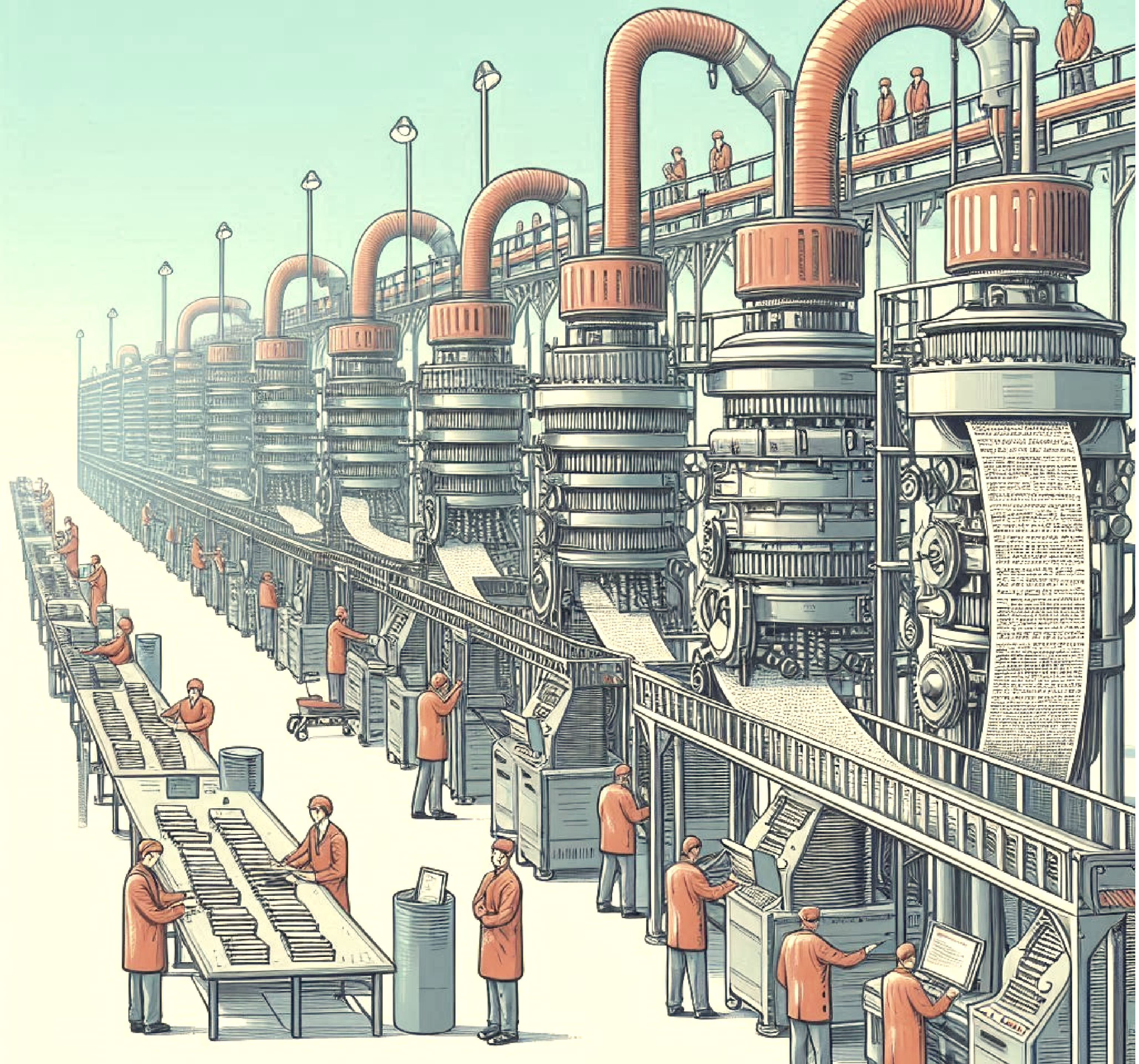
This slide demonstrates a full background image using the `bg` attribute. The image covers the entire slide background.

You can also add effects to the background image, like `blur`, `opacity`, `sepia`, `brightness`, etc. For instance, the image in this slide has the following effects:

```
![bg blur:10px opacity:0.6 sepia:30% brightness:0.9](imgs/example1.jpeg)
```

This slide shows a background image positioned to the right with 40% width and a brightness of 0.6: `bg right:40% brightness:0.6`





This slide demonstrates a background image on the left (60% width) with a sepia filter: `bg left:60% sepia:50%`

Multi-column layouts

Two-column layout

- First column content
- Can contain any element

Important notes:

- When using any html code within the markdown file, **always** remember to leave a blank line before and after the html code, otherwise the markdown will not be rendered correctly.
- Also remember to close all the `<div>` tags that you open, like this: `</div>`.

Second Column

- Additional content here
- Each column is equally sized

Usage:

```
<div class="columns">
<div>

  Content for column 1

</div>
<div>

  Content for column 2

</div>
</div>
```

Three-column layout

Column 1

- Feature A
- Feature B
- Feature C

Column 2

- Feature D
- Feature E
- Feature F

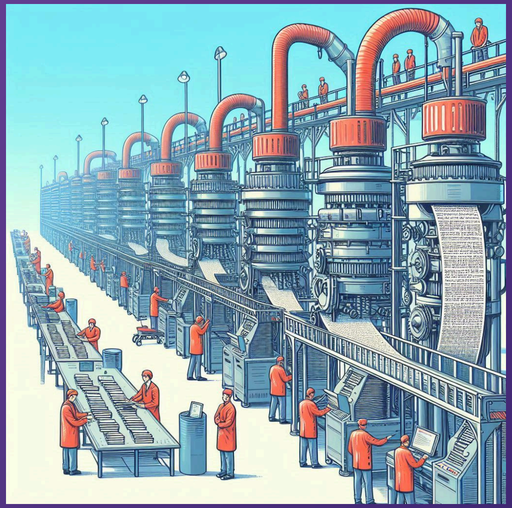
Column 3

- Feature G
- Feature H
- Feature I

We can also adjust the background color of the slides very easily:

```
<!-- _backgroundColor: "lightgreen" -->
<!-- _backgroundColor: "rgb(199, 255, 213)" -->
```

Figure containers with captions



This is a right-aligned figure with a caption.

The figure container allows having floating images with captions.

The text flows naturally around the figure + caption, creating a clean, academic presentation style.

However, the syntax is a bit verbose, so **we try to avoid it if possible**.

Fancy background gradients

```
<!-- _backgroundImage: "linear-gradient(135deg, #667eea 0%, #764ba2 100%)" -->
<!-- _color: white -->
```

Code examples

Python code block (note that we get syntax highlighting for free!)

```python
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.linear_model import LogisticRegression

vect = CountVectorizer()
modelLR = LogisticRegression()

# Train the model
X_train_vectorized = vect.fit_transform(X_train)
modelLR.fit(X_train_vectorized, y_train)

# Make predictions
X_test_vectorized = vect.transform(X_test)
predictions = modelLR.predict(X_test_vectorized)
```

Inline Code

Use `CountVectorizer()` for bag-of-words representation and `TfidfVectorizer()` for TF-IDF weighting.

Tables and data presentation

Algorithm	Accuracy	Precision	Recall	Anything: $\sum_1^N \frac{(y-y_{hat})^2}{N}$
Naive Bayes	0.85	0.83	0.87	0.85
Logistic Regression	**0.88**	0.86	`0.90`	`0.88`
Random Forest	*0.87*	0.85	0.89	0.87

A table can be created with the `|` character. Note that we use the `:` character to customize the alignment of the text in the table.

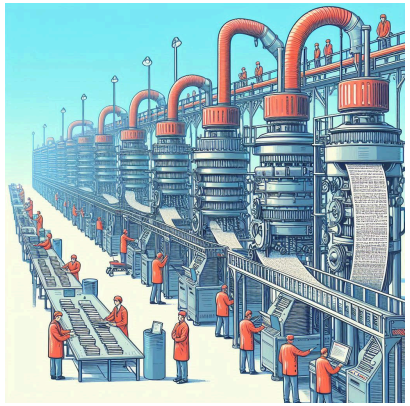
By default, tables are left-aligned within the slide. To center the table (or any other element), you can enclose it in `<div align="center"> ... </div>` tags:

Algorithm	Accuracy	Precision	Recall	Anything: $\sum_1^N \frac{(y-y_{hat})^2}{N}$
Naive Bayes	0.85	0.83	0.87	0.85

Absolute positioning example

This demonstrates absolute positioning capabilities. The image is positioned at specific coordinates (200px from right, 300px from top) with a defined width and height.

Usage Notes:



- Coordinates are relative to the slide
- Default slide size is 1280x720px
- Use z-index to control layering
- **Avoid using absolute positioning if possible**, use the other positioning methods (float, center) instead. This is because it is not responsive, and it will break easily if any elements in the slide are modified (e.g., we change text size).

Reference and citation styling

This slide demonstrates the reference styling feature:

The reference class provides consistent styling for citations and references throughout the presentation.

Natural Language Processing with Python, https://www.nltk.org/book/ - Bird, Klein & Loper

Custom classes and styling

There are several classes you can use to style your slides:

- `<!-- _class: small -->`: This slide uses the `small` class to reduce font size to 24px, perfect for content-heavy slides.
- `<!-- _class: smaller -->`: This slide uses the `smaller` class for even more compact text at 20px.
- `<!-- _class: smallest -->`: This slide uses the `smallest` class for the smallest text at 18px.

Additionally, you can use the `no-footer` class to remove the footer and logos from the slide: `<!-- _class: no-footer -->`. Or combine both like this: `<!-- _class: no-footer smallest -->`.

Mathematical Formulas and Equations

Bayes' Theorem:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Naive Bayes Classification:

$$P(y|x_1, \ldots, x_n) = \frac{P(y) \prod_{i=1}^{n} P(x_i|y)}{P(x_1, \ldots, x_n)}$$

TF-IDF Formula:

$$\text{TF-IDF}(t,d) = \text{TF}(t,d) \times \text{IDF}(t)$$

Where:

- $\text{TF}(t,d)$ is the term frequency
- $\text{IDF}(t) = \log \frac{N}{df(t)}$ is the inverse document frequency

Boxes

Info box

This is an info box. It is used to display important information.

Warning box

This is a warning box. It is used to display warning information.

Error box

This is an error box. It is used to display error information.

Importing material into MARP

Importing images in clipboard (from screenshots, from the web, etc.)

Simply `Ctrl+V` to insert the image into Markdown! To configure the path where the image will be saved, you must edit the `settings.json` file in the `.vscode` folder.

Importing general content from the web

For webpages, you can simply copy stuff from the web and paste it into this webpage to convert it to Markdown: https://euangoddard.github.io/clipboard2markdown/

For better results, consider using the following Chrome extensions:

- For general webpages: WebInk
- For ChatGPT in paritcular (with support for equations): ChatGPT to Markdown Pro Converter

Other tools

To remove the background of an image, you can use the following tool: https://www.iloveimg.com/remove-background

Be as fancy as you want with HTML + CSS ❤️

You can embed any HTML and CSS you want. This allows for creating complex and visually appealing slides that go beyond standard Markdown capabilities.

Instead of the defining a gobal `<style>` tag, you can define a `<style scoped>` tag to make the CSS local to the slide. Here are a few examples:

Modular Design

Component-based architecture.

High Performance

Optimized for low-latency.

Secure by Default

Built-in security features.

Fully Responsive

Adapts to any screen size.